

Oussama Bouhali

Computer Science Engineer | Data Scientist & ML Engineer

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SUMMARY

Computer Science Engineer with a strong background in Data Science and Artificial Intelligence, with a strong interest in data engineering and scalable data solutions. Proficient in Python, SQL, data processing, and database technologies, with hands-on experience working with data pipelines and transforming complex datasets through academic and professional projects. Eager to apply my technical skills to build reliable data pipelines and support data-driven applications.

EDUCATION

ENSI- École Nationale des Sciences de l'Informatique <i>Engineer in Computer Science</i>	Sep 2024 – Jun 2026 Manouba, Tunisia
ISGT – Institut Supérieur de Gestion de Tunis <i>Master in Data Science For Business</i>	Sep 2022 – Jul 2024 Tunis, Tunisia
ISGT – Institut Supérieur de Gestion de Tunis <i>Bachelor's in Business Computing, specializing in Business Intelligence (BI)</i>	Sep 2019 – Jun 2022 Tunis, Tunisia
Menzah 9 High School <i>Baccalaureate in mathematical science (Highest honors with Overall average score 18.29/20)</i>	Sep 2015 – Jun 2019 Tunis, Tunisia

PROFESSIONAL EXPERIENCE

PwC France & Maghreb <i>Machine Learning Intern – Quantitative Finance</i> Modeled SPX 500 market volatility, developed backtesting frameworks, and compared multiple models to assess robustness and predictive performance across different market conditions, including crisis and high-volatility periods. Tools: Python, FastAPI, React.js, TensorFlow, Scikit-learn, PostgreSQL, Docker, Black-Scholes, Monte Carlo	Feb 2026 – Jun 2026 Tunis, Tunisia
EYACOM <i>AI Engineer – Development of SOC Alert Triage and Threat Hunting with Wazuh</i> Developed an AI-based SOC alert triage solution with an automated log-processing and FAISS indexing pipeline, integrating Ollama-Mistral and Conversational Retrieval to enable contextual threat hunting in Wazuh. Tools: Python, Wazuh, FAISS, LangChain, Ollama (Mistral)	Jun 2025 – Aug 2025 Montrouge, France
PwC France & Maghreb <i>Data Scientist</i> Analyzed key features influencing Probability of Default (PD), applied GAN-based resampling to address class imbalance, developed ML models for PD prediction, implemented an internal rating system, and evaluated performance using AUC, precision, and recall Tools: Python, Scikit-Learn, Pandas, numpy, Matplotlib: Seaborn, Plotly, Tensorflow	Feb 2024 – Jun 2024 Tunis, Tunisia
TRIVAW <i>FullStack Developer and Business Intelligence Analyst</i> Implemented rider-driver matching and user feedback systems, and built real-time dashboards with clear data visualizations to optimize the taxi-sharing platform. Tools: React, Nodejs	Feb 2022 – May 2022 Tunis,Tunisia
EDIVOPS <i>Data Engineer and Business Intelligence Analyst</i> Collected and processed historical crime data, engineered features such as crime density and severity, and developed Power BI dashboards to visualize clustered crime patterns through maps, charts, and tables. Tools: ETL, PowerBI	Jul 2021 – Aug 2021 Tunis, Tunisia
BIAT <i>Web Developer</i> Implemented secure user authentication and registration processes, including password- and PIN-based access. Tools: HTML, CSS, JS	Jul 2020 – Aug 2020 Tunis, Tunisia

PROJECTS

Realization of a Context-Aware Speech Recognition for Smart Assistance and Multilingual Interaction <i>Developed a context-aware speech recognition system for smart assistance and multilingual interaction, using deep learning to transcribe and translate speech across different languages while adapting to user context.</i> Tools: PyTorch, Whisper, ResNet, LSTM, LibriSpeech, Common Voice, Mel Spectrograms
Implementation of a Machine Learning project (LinkedIn Data Analyst Jobs Listings) <i>developed a machine learning initiative focused on LinkedIn's data analyst job offers, using powerful classification models aligned with the Crisp-DM methodology (Business Understanding, Data Understanding, Data Cleaning, Data Preparation, Modeling, Evaluation, Deployment)</i> Data Source: Kaggle Tools: Python, Scikit-Learn, Pandas, Matplotlib: Seaborn

Web Application for Iris Species Prediction Using Django and Random Forest

Developed a predictive analytics project using the Django web framework and implemented the Random Forest machine learning algorithm to achieve accurate classification of Iris species based on features

Tools: Python, Django, Scikit-Learn

Realization of a Recommendation System Project

Developed a recommendation system for a website aimed at assisting clients in making purchasing decisions, utilizing collaborative filtering techniques, including user-based, item-based, and matrix factorization methods

Tools: Python, Atoti, PostgreSQL

Compiler Project

Developed a compiler in C with lexical, syntax, and semantic analysis.

Implemented a translator to generate intermediate code.

Tools: C

Data analysis project

Clean data , Analyze data Visualize data using the Dashboard

Tools: Talend Open Studio (ETL), Power BI



COMPUTER SKILLS

Data Visualization — Power BI, Cognos analytics IBM, Atoti, D3.js, Database Management System — Oracle, MYSQL, PostgreSQL, MongoDB, Tools and Software — Android Studio, Netbeans, Talend Open Studio(ETL), Visual Studio Code, Jupyter Notebook, Spss Statistics, Spss Modeler, Machine Learning — Numpy, Pandas, Matplotlib, Scikit-Learn, Tensorflow, LSTM, RNN, GRU, Frameworks — Django, FastAPI, React, Node.js, Methodology — Scrum, CRISP-DM



FINANCIAL RISK MODELING

Backtesting
VaR backtesting (Historical, Monte Carlo), Expected Shortfall backtesting, Kupiec test, Christoffersen test, Basel Traffic Light approach

Credit Risk Modeling
PD, LGD, and EAD modeling; IFRS 9 framework; credit risk calibration, validation, and performance assessment.

Time Series & Market Risk Models
ARIMA, GARCH, volatility modeling, VaR/ES estimation under different distributional assumptions

Derivative Products (Options Pricing)
Understanding of financial derivatives, including call and put options, payoff structures, hedging principles, and introductory option pricing methods such as Black–Scholes, Monte Carlo simulations, and the Binomial model.



LANGUAGES

Arabic Native language	French Professional Proficiency	English Professional Proficiency
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LEADERSHIP EXPERIENCES

Association des Handicapés de Oued Ellil Member	Jul 2018 – Jun 2021 Manouba, Tunisia
Croissant Rouge Tunisien - Comité Régional Ariana Member	Jan 2022 – Apr 2022 Ariana, Tunisia
IEEE ENSI Member	Oct 2023 – present Manouba, Tunisia
Enactus ENSI Member	Oct 2023 – Jun 2024 Manouba, Tunisia
Association Robotique ENSI Member	Oct 2023 – present Manouba, Tunisia
Tuni'act Alpha Member	Oct 2023 – Jan 2025 Tunis, Tunisia



CERTIFICATES

NVIDIA
Building Transformer-Based Natural Language Processing Applications

IEEE Xtreme 17.0 – Certificate of Participation

Participated in the global IEEE Xtreme 17.0 programming competition as part of Team Joker, competing against 17,000+ participants.

Organized by IEEE, October 29, 2023.